

The Cost of Silence: Conditional Under-disclosure, Price Discovery, and the Cross-Section of Stock Returns¹

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Abstract

Under a voluntary ESG disclosure regime with substantial rating disagreement, which end of the disclosure distribution does the market price? The greenwashing literature implicitly assumes that exaggeration is punished. Using 1,339 Chinese A-share firms over 2014–2023, I construct a measure of *conditional under-disclosure*: the within-industry standardized gap between a firm’s ESG disclosure score and its realized ESG performance, orthogonalized each month against six style exposures, industry fixed effects, ESG performance, and fundamentals. What is systematically priced is not exaggeration but *silence*. The orthogonalized gap predicts one-month-ahead returns (IC = 0.0122, $t = 3.45$, defensive timing), and the most silent decile underperforms comparable firms by 4.17% per year (−4.16%, $t = -2.92$, after the [Liu et al. \(2019\)](#) China three-factor adjustment), with near-zero factor loadings. Raw disclosure gaps have *no* predictive power: orthogonalization is a necessary condition for the signal to appear. Three competing explanations are rejected. It is not greenwashing punishment—a governance-dimension placebo is as strong as the environmental dimension, the effect is not amplified in polluting industries, and penalty announcements generate *positive* abnormal returns (+2.76%, $t = 4.98$). It is not a liquidity premium—silent firms are *more* liquid, the mediation share is 7.0%, and the signal survives Amihud neutralization. It is not size or quality. The surviving channel is price discovery: the discount concentrates among stocks with high [Hou and Moskowitz \(2005\)](#) price delay (tail −7.39%, $t = -3.54$, versus −3.01%, $t = -1.57$), and this contrast holds within the low-idiosyncratic- volatility subsample, so it is not a restatement of limits to arbitrage. Excluding the bottom disclosure decile from an equal-weighted portfolio earns 0.47% per year (IR = 1.07), with incremental one-way turnover of 1.06% per month and about nine-tenths of the gain surviving 30bp round-trip costs. Results describe cross-sectional predictability rather than causal effects, the magnitudes are modest, and the effect vanishes under value weighting.

JEL codes: G12, G14, M41, Q56.

Keywords: ESG disclosure, greenwashing, price discovery, information barriers, cross-section of returns.

¹All code, result tables, and negative results are archived and available from the author. All remaining errors are my own.

1 Introduction

China’s ESG market grew under a disclosure regime that was, until very recently, almost entirely voluntary. Firms chose whether and how much to disclose; rating agencies, in turn, leaned heavily on what firms volunteered. The result is an information environment with two structural features: what a firm says is partly a choice variable rather than a fact, and different agencies reach markedly different conclusions about the same firm—in my sample the pairwise rank correlations across three major providers range from 0.26 to 0.69. In April 2024 the three Chinese exchanges issued sustainability reporting guidelines that begin to make disclosure mandatory for index constituents, which makes the pre-mandate period a well-defined laboratory for a simple question: *which end of the disclosure distribution does the market price?*

The existing literature has largely studied the upper end. Work on greenwashing (Yu et al., 2020; Hu et al., 2023) is built on the premise that firms which overstate their environmental credentials are eventually identified and punished. The premise is intuitive, but the cross-sectional pricing evidence for it in China is thin. This paper documents the opposite: *what is systematically priced is not exaggeration but silence.*

The measurement strategy is deliberately simple, and one step in it turns out to matter more than anything else. A firm’s absolute disclosure score is contaminated by size, industry norms, and visibility, so I first standardize disclosure and realized ESG performance within industry and take the difference, producing a disclosure gap. I then, each month, project this gap on six style exposures (size, book-to-market, momentum, beta, idiosyncratic volatility, turnover), industry dummies, and ESG performance, profitability and leverage. The residual—what I call conditional under-disclosure—answers a narrow question: among firms that look alike in size, style, industry, and actual ESG performance, who is relatively silent?

The first result is that this last step is not cosmetic but necessary. The raw disclosure gap has no predictive power whatsoever ($IC = -0.0004$, $t = -0.10$). Style neutralization alone leaves it insignificant (0.0062 , $t = 1.78$). Only after ESG performance and fundamentals are also removed does the signal appear ($IC = 0.0136$, $t = 4.23$; 0.0122 , $t = 3.45$ under the most conservative timing and standardization convention). For comparison, the ESG performance score itself—the variable most ESG-pricing studies use—never predicts returns once styles are removed. This asymmetry is the paper’s methodological point: disclosure behavior carries information that ESG *levels* do not, but only conditionally.

The second result characterizes where the return effect lives. Sorting on the orthogonalized gap produces a one-sided pattern rather than a monotone one: the most silent decile earns 0.264% per month against a universe mean of 0.612%, while deciles two through ten are tightly clustered. Relative to comparable firms the silent decile loses 4.17% per year ($t = -3.14$), and -4.16%

($t = -2.92$) after adjusting for the [Liu et al. \(2019\)](#) China three-factor model, with essentially zero loadings on market, size, and value.

The third set of results rejects three competing explanations. *It is not greenwashing punishment.* If the market were pricing green window-dressing, the effect should be concentrated in the environmental dimension; instead a governance-dimension placebo is at least as strong (IC = 0.0120, $t = 3.62$, versus 0.0099, $t = 3.37$) even though the two component factors correlate only 0.18. The effect is not amplified in heavily polluting industries. And in an event study around 2,094 hand-collected penalty announcements, high-gap firms earn *positive* cumulative abnormal returns of +2.76% ($t = 4.98$)—the opposite sign to what “exposure of greenwashing” predicts, and consistent with Chinese penalty filings typically being published jointly with confirmation of remediation, so that the filing resolves uncertainty rather than creating it. *It is not a liquidity premium.* Silent firms are, if anything, *more* liquid conditional on size and turnover; the liquidity channel mediates only 7.0% of the total effect; and adding log-Amihud illiquidity to the neutralization leaves predictive power essentially intact (0.0136 $\rightarrow$ 0.0128, $t = 3.90$). *It is not size or quality:* the effect is stronger in the large-cap half, and the signal is orthogonal to a standard quality composite (correlation 0.001) while retaining a Fama-MacBeth marginal t of 3.68 on top of it.

The fourth result identifies the channel that survives. Using [Hou and Moskowitz \(2005\)](#) price delay, the silence discount concentrates precisely where prices absorb market information slowly: in the high-delay half the most silent decile loses 7.39% per year ($t = -3.54$) versus 3.01% ($t = -1.57$) in the low-delay half. Because delay correlates with idiosyncratic volatility ($\rho = 0.37$), and high idiosyncratic volatility is the standard proxy for limits to arbitrage, I repeat the split inside the low-volatility subsample; delay still separates the two groups (−5.94%, $t = -3.11$, versus −3.43%, $t = -1.93$). Price-discovery speed therefore carries information beyond arbitrage frictions. Together with the event study, the picture is a barrier that is raised, priced while it stands, and repriced when it is removed.

Finally, the economic magnitudes are real but small, and I report them as such. Excluding the bottom disclosure decile from an equal-weighted portfolio generates 0.47% per year with a tracking error of 0.45% (IR = 1.07). Because the exclusion list covers only a tenth of the portfolio, incremental one-way turnover is 1.06% per month and roughly nine-tenths of the gain survives 30bp round-trip costs. Under value weighting the excess return disappears, so the economic relevance is confined to equal-weighted or small-cap-tilted implementations.

What this paper does and does not claim. This is a study of cross-sectional *predictability*, not of causal effects. Disclosure is a choice, and the neutralization controls observables only. I attempted to use the April 2024 disclosure mandate as a quasi-natural experiment and report in Section 9 why that test cannot identify anything in the current data: the mandate was announced the same day as a major market-structure policy that triggered a large-cap/small-cap divergence, my treatment

proxy is constructed from market capitalization and thus overlaps that shock, and the event window coincides with a period in which the baseline effect had itself reversed. I keep the negative result rather than discarding it.

Related literature. This paper connects three strands. The first is the greenwashing literature (Yu et al., 2020; Hu et al., 2023), whose maintained hypothesis—that overstatement is punished—I test directly and reject in the Chinese cross-section, while confirming Hu et al. (2023)’s “green fog” intuition in a different form: the tail discount is deepest where rating agencies disagree most. The second is the asset-pricing literature on incomplete information: Merton (1987) on investor recognition and Easley and O’Hara (2004) on information risk both imply that information barriers should be priced, though neither pins down whether the consequence is compensation for risk or slow-moving mispricing; my evidence favors the latter. The third is the price-discovery literature, in particular Hou and Moskowitz (2005), whose delay measure provides the channel test that discriminates between the two mechanisms considered here.

2 Institutional setting and hypotheses

2.1 Setting

Before the April 2024 guidelines, ESG reporting by Chinese listed firms was voluntary apart from narrow sector rules; by the end of 2024 roughly 2,200 listed firms published an ESG or CSR report, with wide variation in quality and comparability. Two consequences matter here. First, disclosure quantity is itself an input to ratings, so “discloses a lot” and “performs well” are hard to separate within any single rating. Second, agencies disagree: the pairwise rank correlations among the three providers used in Section 7 are 0.685, 0.453, and 0.255.

2.2 Hypotheses

Merton (1987) shows that securities investors do not know about require higher expected returns; Easley and O’Hara (2004) shows that securities with greater information asymmetry command higher required returns. Both imply that information barriers are priced, but the direction of the return effect depends on whether the barrier operates as compensated risk or as slow-moving mispricing. This motivates:

H1 (silence discount). Conditional on size, style, industry, and realized ESG performance, relatively under-disclosing firms earn lower future returns.

H2 (liquidity channel). Under-disclosing firms are less liquid, and liquidity differences account for a substantial part of the discount.

H3 (price-discovery channel). Under-disclosing firms exhibit higher price delay, and the discount is concentrated among firms whose prices adjust slowly.

H4 (barrier removal). When information is released by regulatory compulsion, firms with previously higher barriers experience a significant price reaction.

The greenwashing literature’s implicit hypothesis—that *high* disclosure gaps predict low returns—is the mirror image of H1 and is treated throughout as the competing hypothesis.

3 Data and measurement

3.1 Sample

ESG disclosure scores are Bloomberg ESG disclosure scores (0–100, measuring breadth and granularity of disclosure); ESG performance is the numerical rating of a major domestic Chinese ESG rating provider.² Accounting data are from CSMAR. After merging and cleaning, the annual panel covers 2014–2023 with 9,835 firm-year observations on 1,339 firms in 71 industries. Daily prices for the entire A-share market (January 2014 to December 2024) are used to build monthly returns, style factors, and the liquidity and price-discovery measures. Merging the annual signal with monthly returns yields 120 monthly cross-sections from January 2015 to December 2024 (119 with forward returns), 116,244 stock-months, and 969 stocks per month on average.

3.2 Conditional under-disclosure

The disclosure gap is

$$GWS_{i,t} = z(Disclosure_{i,t} | Ind_i) - z(Performance_{i,t} | Ind_i), \quad (1)$$

where $z(\cdot | Ind)$ denotes within-industry cross-sectional standardization. Low values identify firms that do relatively more than they say. Each month I then estimate

$$GWS_{i,t} = \alpha_t + \sum_k \beta_{k,t} Style_{k,i,t} + \sum_j \gamma_{j,t} Ind_{j,i} + \delta'_t Fund_{i,t} + \varepsilon_{i,t}, \quad (2)$$

where *Style* contains log market capitalization, book-to-market, momentum (cumulative return from $t - 12$ to $t - 2$), beta (250 trading days against the CSI All-Share index), idiosyncratic volatility, and

²The exact provider should be confirmed against the data manual before submission; the empirical design requires only that the performance leg be a rating not mechanically driven by disclosure quantity.

Table 1: Cross-sectional predictive power by level of processing. IC is the Pearson correlation between the signal and the next month’s return computed within each monthly cross-section; RankIC is its Spearman analogue. ICIR is the mean IC divided by its standard deviation across months, and t multiplies the ICIR by $\sqrt{T}$. Baseline timing uses $T = 119$ months (2015:01–2024:12); defensive timing uses $T = 113$.

Signal	IC	ICIR	$t(\text{IC})$	RankIC	$t(\text{RankIC})$	% pos.
<i>GWS</i> raw	−0.0004	−0.01	−0.10	−0.0052	−1.08	48.7
+ style	+0.0062	0.16	1.78	+0.0037	1.06	54.6
+ style + ESG/fundamentals	+0.0136	0.39	4.23	+0.0124	3.45	64.7
ESG performance raw	+0.0034	0.04	0.48	+0.0096	1.26	46.2
+ style	+0.0046	0.11	1.21	+0.0052	1.48	56.3
Defensive timing (primary)	+0.0122	0.32	3.45	+0.0103	2.87	62.8

turnover, each standardized and winsorized at ± 3 ; *Fund* contains the ESG performance score, ROA, and leverage. The residual $GWS^\perp$ is the paper’s explanatory variable.

3.3 Timing

Year- t ESG scores carry an accounting period ending 31 December of year t , so they are computed from annual reports that are published between April and June of $t + 1$. I therefore report two conventions. Under *baseline timing* the year- t signal is applied to the twelve monthly cross-sections of calendar year $t + 1$. Under *defensive timing*—the paper’s primary specification—it is applied from July of $t + 1$ for the following twelve months, and industry standardization is replaced by industry-by-year standardization so that no standardization constant is reused across years. Tables report both.

4 The silence discount

4.1 Orthogonalization is necessary

Table 1 and Figure 1 report predictive power at three levels of processing. The raw gap predicts nothing; style neutralization alone is insufficient; only after ESG performance and fundamentals are removed does the signal emerge. The ESG performance score itself, by contrast, never survives style neutralization. This is the central measurement result: what is informative is disclosure *conditional* on everything that ordinarily drives it.

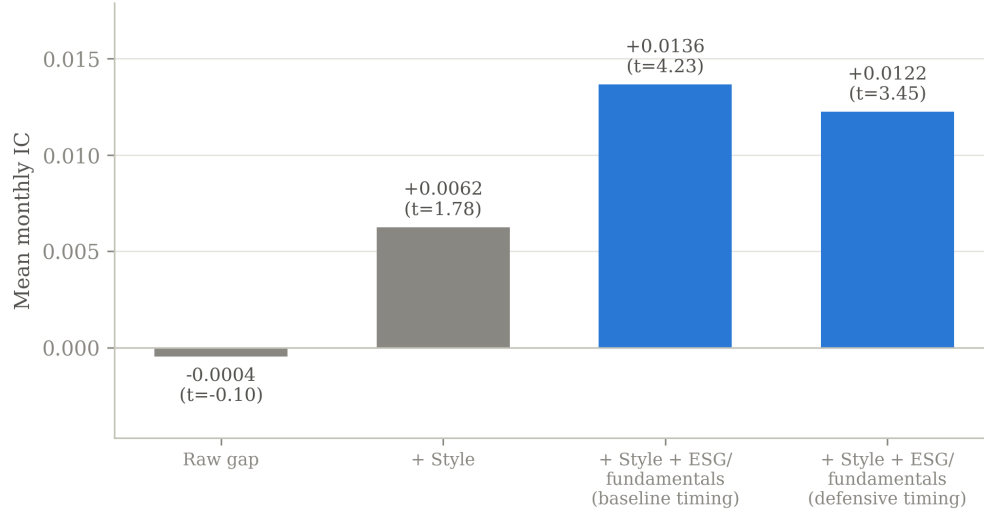


Figure 1: Orthogonalization steps and predictive power. Bars are mean monthly ICs with t -statistics in parentheses. Grey bars denote incompletely neutralized specifications.

Table 2: Average next-month returns by conditional under-disclosure decile. Defensive timing. D1 contains the most silent firms. The long-short portfolio (D10 – D1) earns 4.44% per year with a Sharpe ratio of 0.79 ($t = 2.42$) and a maximum drawdown of -10.25% .

	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Universe
Return (%)	0.264	0.407	0.606	0.637	0.651	0.661	0.627	0.788	0.847	0.634	0.612

4.2 Portfolio sorts

Table 2 and Figure 2 show that the effect is one-sided. The most silent decile earns 0.264% per month against a universe mean of 0.612%; deciles two through ten differ little from one another. The long-short portfolio earns 4.44% per year with a Sharpe ratio of 0.79 ($t = 2.42$), and the silent decile loses 4.17% per year relative to comparable firms ($t = -3.14$). Figure 3 plots cumulative values and Figure 4 the monthly RankIC series. The one-sidedness has a direct implication for implementation, developed in Section 8: the signal should be used as an exclusion flag rather than as a linear score.

4.3 Risk-adjusted alphas

Table 3 regresses portfolio returns on the CAPM, on Fama-French three factors, and on the China three factors of [Liu et al. \(2019\)](#), which I construct from the same daily data: the smallest 30% of firms by market capitalization are dropped to avoid shell-value contamination, value is measured by earnings-to-price, and the six 2×3 portfolios are value weighted. Over the sample the market, size,

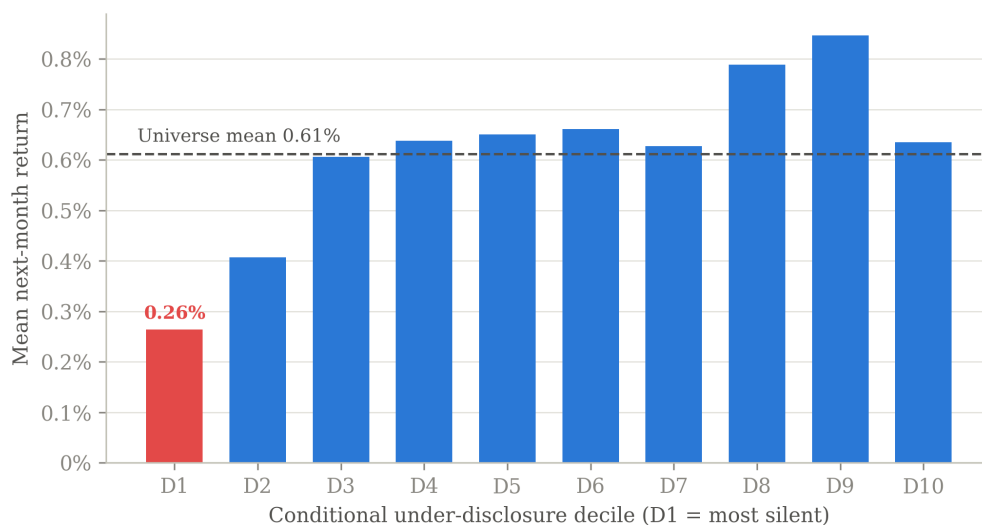


Figure 2: Average next-month returns by decile (defensive timing).

and value factors earn 7.14%, 3.19%, and 9.53% per year. Standard errors follow [Newey and West \(1987\)](#) with six lags.

Two features deserve emphasis. First, the silent-decile discount and the exclusion portfolio's gain are significant at the 1% level under every model, whereas the long-short alpha under the most conservative combination (defensive timing, Fama-French three factors) reaches only $t = 1.78$ —a pattern consistent with the information residing in the lower tail. Second, the "silent decile" rows measure performance *relative to the equal-weighted coverage universe*. In absolute terms the silent decile's China three-factor alpha is only -0.25% ($t = -0.18$) while the top decile earns 3.62% ($t = 2.21$), because the coverage universe—Bloomberg-covered and therefore large-cap tilted—itself carries a positive alpha against the three-factor model. The correct statement is that silent firms are discounted relative to comparable firms, not that they lose money in absolute terms.

5 What the effect is not

5.1 Not greenwashing punishment: a governance placebo

If the market priced green window-dressing, the effect should be specific to the environmental dimension. Table 4 constructs separate gaps from Bloomberg's E, S, and G disclosure sub-scores. The governance gap is at least as predictive as the environmental gap even though the two component factors correlate only 0.183; an equal-weighted composite of the three adds nothing over the aggregate gap (IC = 0.0132, $t = 4.18$). Two nearly orthogonal signals with equal predictive power indicate a common component—disclosure behavior across dimensions—rather than anything

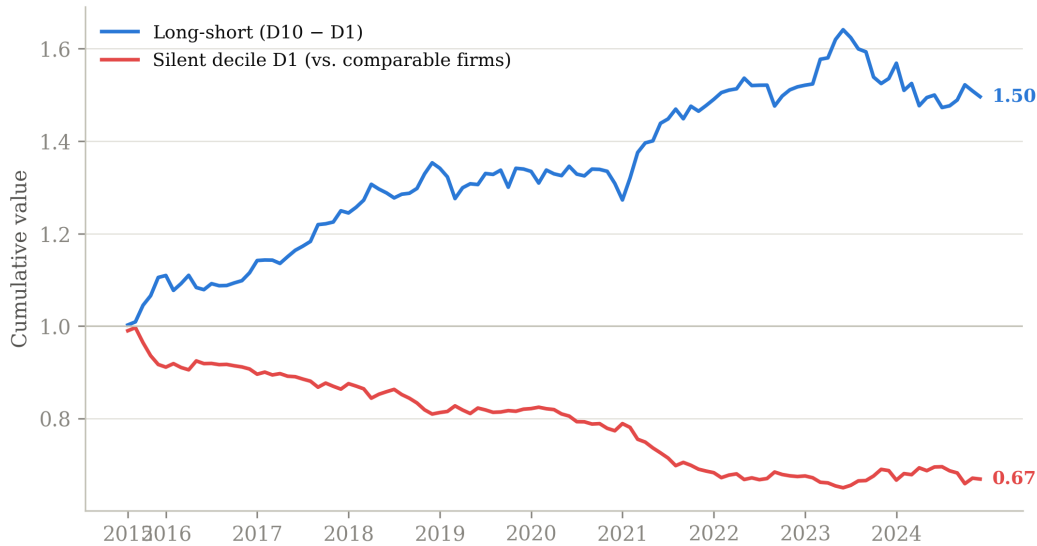


Figure 3: Cumulative value of the long-short portfolio and of the silent decile relative to comparable firms (defensive timing).

specific to environmental claims.

5.2 Not greenwashing punishment: polluting industries

Incentives to greenwash are strongest in heavily polluting industries. Using three alternative definitions of "heavy polluter," Table 5 finds no consistent amplification: the effect is marginally stronger among polluters under the first definition and insignificant among them under the third.

5.3 Not greenwashing punishment: penalty announcements

If exaggeration were punished upon exposure, penalty filings should produce negative abnormal returns, especially for high-gap firms. I collect all company filings from the CNINFO disclosure portal whose titles contain the word "penalty" between 2014 and 2024: 2,094 filings by 921 firms, of which 1,561 concern securities regulation and 73 concern environmental violations. Table 6 and Figure 5 report market-adjusted cumulative abnormal returns by pre-announcement disclosure gap.

The sign is the opposite of the greenwashing prediction. A reading consistent with Chinese filing practice is that penalty announcements are typically published together with confirmation that remediation is complete (titles routinely read "notice of receipt of administrative penalty decision and completion of rectification"), so the filing removes uncertainty; the information barrier falls, and the revaluation dominates the negative content of the penalty itself.

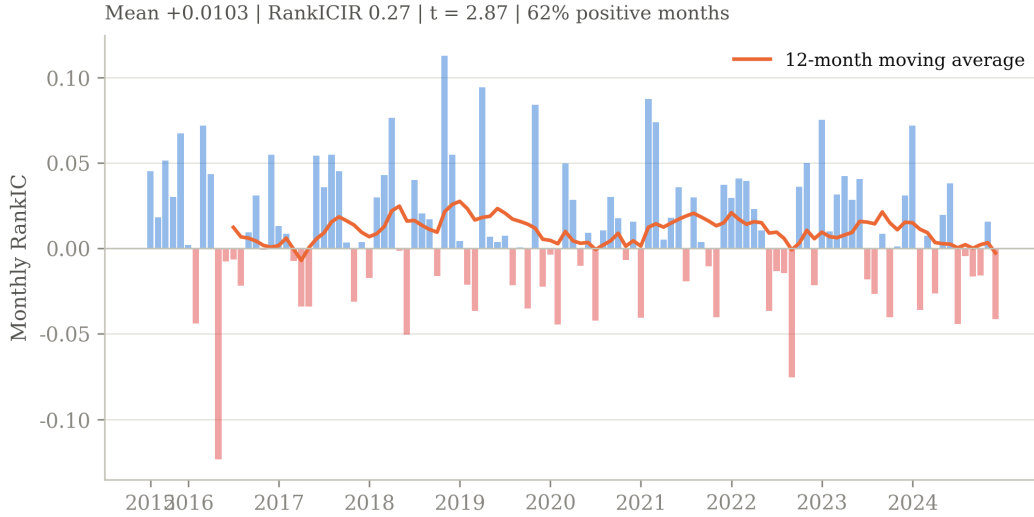


Figure 4: Monthly RankIC and its twelve-month moving average (defensive timing).

5.4 Not an ESG level, quality, or size effect

Table 1 already shows that ESG performance does not survive style neutralization. For quality, I build $Q = z(ROA) - z(Leverage) - z(IdioVol)$; its correlation with $GWS^\perp$ is 0.001. Adding the exclusion list on top of a portfolio of the top 30% of Q (dropping 29 stocks per month on average) contributes 0.49% per year with an information ratio of 0.67 ($t = 2.10$), and in Fama-MacBeth regressions the marginal t -statistic on $GWS^\perp$ is 3.68 against 0.68 for Q itself (Table 7). For size, the double sort in Table 9 shows the effect is *stronger* in the large-cap half (RankIC = 0.0160, $t = 3.63$) than in the small-cap half (0.0080, $t = 1.57$), which rules out a micro-cap junk interpretation; the disappearance of the effect under value weighting (Section 8) is a weighting mechanic, since excluded names carry very small value weights.

6 Mechanism: liquidity or price discovery?

6.1 The liquidity channel is rejected

Table 8 reports three tests of H2 that point the same way. First, the sign is wrong for the hypothesis: Fama-MacBeth regressions of $\log$ Amihud (2002) illiquidity on $GWS^\perp$ give a *positive* coefficient (0.0407, $t = 15.69$; 0.0414, $t = 16.08$ with size and turnover controls), meaning relatively silent firms are more liquid. Second, the mediated share is small: the indirect effect is 7.0% of the total (Sobel 1982 $z = 2.34$, $p = 0.019$), and its sign corresponds to the conventional illiquidity premium rather than to a barrier-induced risk premium. Third, the signal survives: adding $\log$ -Amihud to the neutralization moves the IC only from 0.0136 to 0.0128 ($t = 3.90$), and adding price delay as well

Table 3: Risk-adjusted alphas. Annualized intercepts from time-series regressions of monthly portfolio returns on the CAPM, the Fama-French three factors, and the [Liu et al. \(2019\)](#) China three factors (CH-3). The main panel uses defensive timing ($T = 113$); the bottom panel repeats the exercise under baseline timing ($T = 119$). Standard errors are [Newey and West \(1987\)](#) with six lags. Portfolios in the first three blocks are hedged or relative and are not adjusted for the risk-free rate.

Portfolio	Model	α (ann.)	t	β_{MKT}	β_{SMB}	β_{VAL}
Silent decile D1 (vs. universe)	CAPM	-4.17%	-3.04	0.00		
	CH-3	-4.16%	-2.92	0.02	-0.11	0.00
	FF3	-3.42%	-2.65	0.01	-0.10	-0.02
Exclusion portfolio (drop bottom 10%)	CAPM	+0.47%	3.03	0.00		
	CH-3	+0.46%	2.91	0.00	0.01	0.00
	FF3	+0.38%	2.65	0.00	0.01	0.00
Long-short (D10 – D1)	CAPM	+4.49%	2.59	-0.03		
	CH-3	+3.88%	2.08	-0.03	0.15	0.05
	FF3	+3.27%	1.78	-0.03	0.13	0.06
<i>Baseline timing (comparison)</i>						
Silent decile D1	CH-3	-4.63%	-4.00	0.01	-0.17	-0.01
Exclusion portfolio	CH-3	+0.52%	4.00	0.00	0.02	0.00
Long-short	CH-3	+5.30%	3.38	-0.01	0.07	0.04

leaves it unchanged. The silence discount is therefore not a liquidity premium in disguise.

6.2 The price-discovery channel survives

I measure price delay following [Hou and Moskowitz \(2005\)](#): using 52 weeks of weekly returns, I regress each stock on the contemporaneous market return and then on the contemporaneous return plus four weekly lags, and define $DELAY = 1 - R^2_{\text{restricted}}/R^2_{\text{unrestricted}}$. Higher values mean prices absorb market information more slowly. In my sample the measure averages 0.27 with a median of 0.19 and declines monotonically with size (from 0.304 in the smallest decile to 0.250 in the largest), as in the original study.

Direct evidence is weak: silent firms have marginally higher delay (Fama-MacBeth coefficient -0.0016 , $t = -1.71$). The conditional evidence is strong. Table 9 and Figure 6 show the discount is concentrated where price discovery is slow, and the contrast across delay groups is larger than across any other environment variable considered.

6.3 Discriminating price discovery from limits to arbitrage

Price delay correlates with idiosyncratic volatility ($\rho = 0.37$), the standard proxy for limits to arbitrage, so the previous result could be a restatement of that friction. Table 10 repeats the delay

Table 4: E/S/G sub-dimension gaps: a placebo test. Each row constructs the gap using the corresponding Bloomberg disclosure sub-score in place of the aggregate. Pairwise correlations of the resulting factors are 0.183 (E,G), 0.475 (E,S), and 0.120 (S,G). Baseline timing, $T = 119$.

Signal	IC	$t(\text{IC})$	RankIC	$t(\text{RankIC})$
Aggregate gap	+0.0136	4.23	+0.0124	3.45
Environmental (E) gap	+0.0099	3.37	+0.0067	1.90
Social (S) gap	+0.0066	2.27	+0.0050	1.55
Governance (G) gap (<i>placebo</i>)	+0.0120	3.62	+0.0094	2.75

Table 5: Heavy-polluting industries. Three alternative industry classifications. "Tail excess" is the annualized return of the most silent 10% of the subsample relative to the subsample mean. Baseline timing.

Definition	Subsample	Stocks/month	RankIC	t	Tail excess	t
(1)	Polluting	234	+0.0173	2.70	−5.70%	−2.24
	Other	662	+0.0113	2.92	−5.20%	−3.22
(2)	Polluting	310	+0.0134	2.52	−4.98%	−2.26
	Other	586	+0.0122	3.05	−5.56%	−3.46
(3)	Polluting	143	+0.0047	0.52	−1.57%	−0.47
	Other	753	+0.0141	4.15	−5.62%	−4.06

split *within* the low-volatility half of the sample. Delay continues to separate the two groups, indicating that price-discovery speed carries information beyond arbitrage costs.

Section 5.3 completes the picture. Penalty filings act as forced information releases—moments when the barrier is removed—and firms with the highest prior gaps earn positive abnormal returns as uncertainty resolves. The mechanism the evidence supports is therefore: a barrier is raised, prices adjust more slowly, a discount accumulates, and the discount is repaired when the barrier falls.

7 Heterogeneity: rating disagreement and index membership

[Hu et al. \(2023\)](#) argue that disagreement among raters indicates a murky ESG information environment. I measure disagreement as the cross-provider standard deviation of percentile ranks from three providers (5,142 firms have at least two ratings) and split at the monthly median. Table 11 shows an informative asymmetry: ranking information is stronger where disagreement is low, whereas the tail discount is concentrated where disagreement is high. This is consistent with the green-fog view—mispricing of the most silent firms is deepest precisely where professional assessors cannot agree.

An important caveat: the three rating snapshots are current (August 2026), i.e. after the sample period, so this test relies on disagreement being a persistent firm attribute. I treat it as suggestive

Table 6: Event study around penalty announcements. Market-adjusted cumulative abnormal returns relative to the CSI All-Share index. Firms are split by the previous month's GWS^L against that month's median.

Group	Events	CAR $[-5, +20]$	t	CAR $[0, +5]$	% negative
High disclosure gap	695	+2.76%	4.98	+1.04%	46.5
Low disclosure gap	655	+0.53%	1.04	-0.15%	50.8

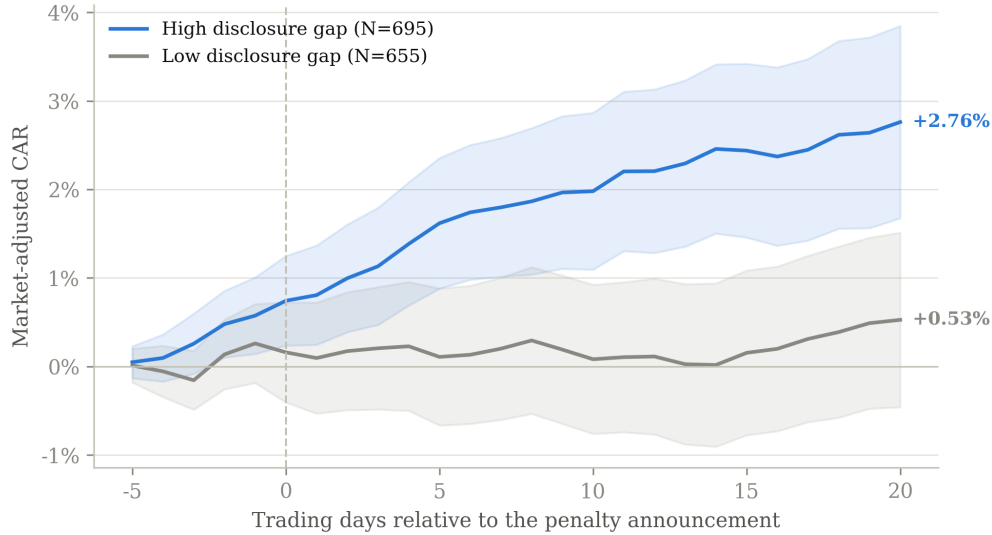


Figure 5: Cumulative abnormal returns around penalty announcements. Shaded bands are 95% confidence intervals. Baseline timing.

rather than conclusive and have begun archiving monthly snapshots so that the test can be redone with point-in-time data.

8 Economic significance

Because the information is concentrated in the lower tail, the natural implementation is an exclusion flag rather than a linear score: drop the most silent 10% (89 stocks per month on average) from an equal-weighted portfolio. Table 12 reports the results.

Three points deserve emphasis. First, costs are not binding: since the list covers only a tenth of the portfolio, incremental one-way turnover is 1.06% per month and about nine-tenths of the gain survives 30bp round-trip costs. Second, a once-a-year list is viable—freezing the list annually (list turnover 6.3% per month) still yields 0.41% per year ($t = 2.43$), which places almost no demand on execution. Third, capacity is genuinely limited: under value weighting the excess return disappears, because excluded names carry small value weights. The economic content of this signal is confined

Table 7: Marginal contribution on top of a quality composite. $Q = z(ROA) - z(Leverage) - z(IdioVol)$; the base portfolio is the top 30% of Q , equal weighted. Excess returns are measured against the equal-weighted universe.

	Quality base	Base + exclusion list	Marginal
Annualized excess	1.67%	2.17%	0.49%
Information ratio	0.30	0.40	0.67
t	0.93	1.26	2.10

Table 8: The liquidity channel. Fama-MacBeth cross-sectional regressions and a three-step mediation test on monthly cross-sections. The survival rows re-estimate the signal with the indicated variable added to the monthly orthogonalization.

Test	Estimate	t / p
a path: $GWS^\perp \rightarrow \ln(Amihud)$	0.0407	$t = 15.69$
b path: $\ln(Amihud) \rightarrow \text{return (controlling } GWS^\perp)$	0.00356	$t = 2.36$
c : total effect	0.00207	$t = 3.87$
c' : direct effect	0.00194	$t = 3.69$
Indirect effect $a \times b$ (mediated share)	0.000145 (7.0%)	$z = 2.34, p = 0.019$
IC after adding $\ln(Amihud)$ to neutralization	0.0128	$t = 3.90$
IC after adding $\ln(Amihud)$ and delay	0.0128	$t = 3.91$

to equal-weighted or small-cap-tilted mandates.

9 Robustness and a failed identification attempt

Timing. Moving the application date from January to July of $t + 1$ lowers the RankIC from 0.0124 ($t = 3.45$) to 0.0103 ($t = 2.87$) and the portfolio gain from 0.57% to 0.47%—an attenuation of roughly 15% with no change in conclusions.

Standardization. Replacing industry standardization with industry-by-year standardization, which removes any cross-year reuse of standardization constants, leaves the results unchanged (it is part of the defensive specification).

Signal age. Splitting months by how long the annual signal has been in use, the RankIC is 0.0123 ($t = 2.30$) in the first six months and 0.0125 ($t = 2.58$) in the last six. The signal captures a slow-moving firm attribute rather than news, and annual updating costs nothing.

Alternative construction. A signal built directly from the residualized disclosure score, without forming a gap at all, correlates 0.971 with $GWS^\perp$ and performs almost identically (IC = 0.0142, $t = 4.21$). After full neutralization the "disclosure minus performance" gap is effectively a measure of *conditional disclosure*; the performance leg mainly serves an orthogonalizing role.

Table 9: Where the silence discount is strongest. Each panel splits the sample at the monthly median of the indicated variable. "Tail excess" is the annualized return of the most silent 10% within the subsample relative to the subsample mean. Baseline timing.

Split variable	Subsample	RankIC	t	Tail excess	t
Price delay	High (barrier high)	+0.0168	3.57	−7.39%	−3.54
	Low	+0.0082	1.66	−3.01%	−1.57
Idiosyncratic volatility	High	+0.0164	3.25	−6.34%	−2.90
	Low	+0.0103	2.13	−4.66%	−3.20
Illiquidity	High	+0.0089	1.84	−6.38%	−3.72
	Low	+0.0136	3.24	−3.11%	−1.78
Market capitalization	High	+0.0160	3.63	−5.11%	−3.18
	Low	+0.0080	1.57	−2.96%	−1.60

Table 10: Delay split within the low-idiosyncratic-volatility subsample. The sample is first restricted to stocks below the monthly median of idiosyncratic volatility and then split at the median of price delay.

Subsample	Stocks/month	RankIC	t	Tail excess	t
Low volatility $\times$ high delay	233	+0.0160	2.37	−5.94%	−3.11
Low volatility $\times$ low delay	233	+0.0085	1.34	−3.43%	−1.93

Real outcomes. In the annual firm panel with industry and year fixed effects and clustered standard errors, the disclosure gap enters ROA with a coefficient of 0.0111 ($t = 3.73$) once ESG performance is controlled (and -0.0042 , $t = -2.23$, without it—a textbook suppression pattern), and the result is unchanged using one-year-ahead ROA or 1%/99% winsorization.

A failed identification attempt. I attempted to use the April 2024 sustainability-reporting guidelines as an exogenous removal of the information barrier in a difference-in-differences design, and abandoned the test for three reasons. First, China’s revised "Nine Guidelines" for capital market development were released on *the same day* (12 April 2024) and triggered a sharp large-cap/small-cap divergence; my treatment proxy is built from market capitalization and therefore overlaps the beneficiaries of that shock, so the two cannot be separated. Second, the event window coincides with a period in which the baseline effect had itself reversed sign, leaving no discount whose narrowing could be detected. Third, the triple-difference estimate is insignificant (2.2% annualized, $t = 0.21$) and size-stratified estimates swing between -49% and $+38\%$. I report this as a negative result rather than dropping it; the natural replacement is to wait until mandatory reports for fiscal 2025 are filed and use *observed changes in disclosure* instead of membership proxies.

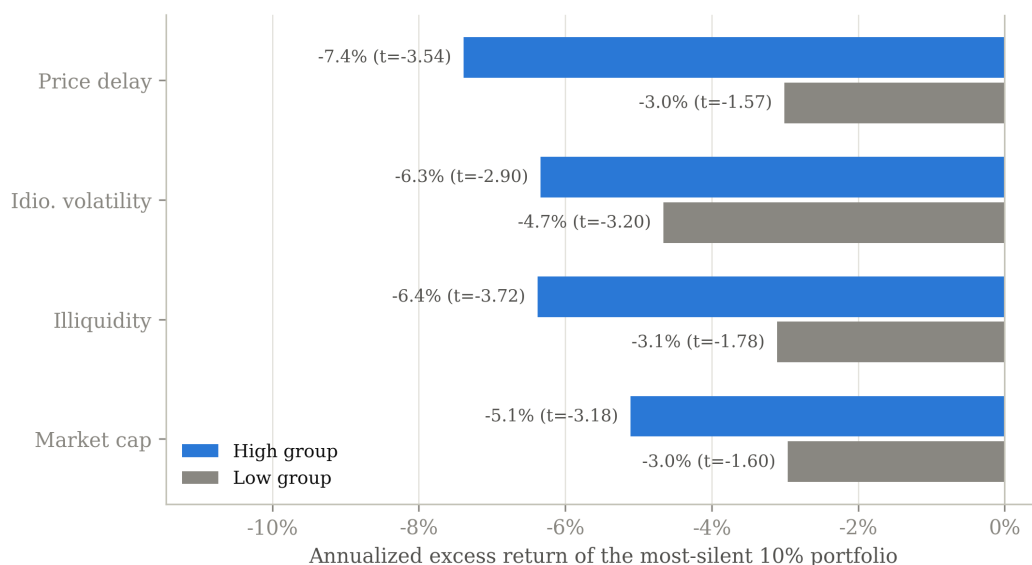


Figure 6: Where the silence discount is strongest. Annualized excess return of the most-silent decile within each subsample, with t -statistics.

Table 11: Rating disagreement and index membership. Disagreement is the cross-provider standard deviation of ESG percentile ranks; index membership uses month-end constituent lists. Tail excess is defined as in Table 9.

Dimension	Subsample	Stocks/month	RankIC	t	Tail excess	t
Rating disagreement	Low	445	+0.0177	4.35	-3.81%	-1.76
	High	444	+0.0088	1.72	-6.34%	-3.93
Index membership	CSI 300	191	+0.0137	2.26	-2.56%	-1.14
	CSI 500	321	+0.0137	2.44	-4.83%	-2.89
	Non-index	384	+0.0096	1.69	-6.83%	-3.14

10 Conclusion

Among Chinese listed firms that look alike in size, style, industry, and realized ESG performance, those that disclose relatively little earn lower subsequent returns—about 4.2% per year for the bottom decile relative to comparable firms, robust to the CAPM, Fama-French, and China three-factor adjustments. The effect is not the mirror image of greenwashing punishment: a governance placebo is equally strong, polluting industries show no amplification, and penalty announcements produce positive abnormal returns. It is not a liquidity premium: silent firms are more liquid and the mediated share is 7%. The channel that survives is price discovery—the discount concentrates where prices adjust slowly, and this is not merely a limits-to- arbitrage effect.

Three limitations bound these conclusions. This is a predictability study, not a causal one:

Table 12: Portfolio implementation. Equal-weighted, monthly rebalanced, baseline timing. Excess returns are measured against the equal-weighted coverage universe. Under defensive timing the monthly-refresh variant earns 0.47% per year with an information ratio of 1.07 ($t = 3.28$).

	Monthly list	Annual frozen list	After 30bp	Value weighted
Annualized excess	0.57%	0.41%	0.53%	−0.19%
Tracking error	0.49%	0.53%	0.49%	1.30%
Information ratio	1.17	0.77	1.09	−0.15
t	3.67	2.43	3.44	−0.47
% positive months	67.2	63.0	66.4	51.3
Profit factor	2.43	1.82	—	—
Max relative drawdown	−0.44%	−0.68%	—	—
List turnover (monthly)	17.9%	6.3%	—	—
Incremental one-way turnover	1.06%	n.a.	1.06%	—

disclosure is chosen, and neutralization controls observables only. The magnitudes are modest—a monthly IC around 0.012 and a portfolio gain around half a percent per year—so the natural use of the signal is as a low-correlation component or exclusion list within a broader model, not as a standalone strategy. And coverage is limited to the roughly 970 Bloomberg-rated firms per month, about a fifth of the market and tilted large, so extrapolation to uncovered firms is unwarranted. A twelve-month rolling RankIC turned negative after the second half of 2023 and the long-short portfolio lost 1.7% in 2024, plausibly as tightening disclosure rules compressed the very dispersion this paper exploits; a twenty-four-month window has not yet turned, so the sample cannot yet say whether the effect has decayed permanently.

Three extensions follow directly. Broader ESG coverage would allow the extensive margin—firms that disclose nothing at all—to be studied, which is the most extreme form of the silence documented here. Once fiscal-2025 mandatory reports are filed, observed disclosure changes offer a cleaner identification strategy than the one that failed here. And textual measures of disclosure specificity and promise fulfillment would separate “silence” from “empty talk,” a distinction this paper’s score-based measure cannot make.

References

- Yakov Amihud. Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1):31–56, 2002.
- David Easley and Maureen O’Hara. Information and the cost of capital. *Journal of Finance*, 59(4): 1553–1583, 2004.
- Kewei Hou and Tobias J. Moskowitz. Market frictions, price delay, and the cross-section of expected returns. *Review of Financial Studies*, 18(3):981–1020, 2005.
- Xin Hu, Renhai Hua, Qingfu Liu, and Chuanjie Wang. The green fog: Environmental rating disagreement and corporate greenwashing. *Pacific-Basin Finance Journal*, 78:101952, 2023.
- Jianan Liu, Robert F. Stambaugh, and Yu Yuan. Size and value in china. *Journal of Financial Economics*, 134(1):48–69, 2019.
- Robert C. Merton. A simple model of capital market equilibrium with incomplete information. *Journal of Finance*, 42(3):483–510, 1987.
- Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.
- Michael E. Sobel. Asymptotic confidence intervals for indirect effects in structural equation models. *Sociological Methodology*, 13:290–312, 1982.
- Ellen Pei-yi Yu, Bac Van Luu, and Catherine Huirong Chen. Greenwashing in environmental, social and governance disclosures. *Research in International Business and Finance*, 52:101192, 2020.